Data-Driven Models for Reducing Food Waste in Supply Chains

Data Source: https://www.kaggle.com/datasets/joebeachcapital/food-waste

Student Name:

Student ID:

Supervisor Name:

Acknowledgement

I would like to officially acknowledge my academic supervisor because he was my source of great advice, great feedback, and support all through this research. It is also observed that appreciation is given to the institution as it offers the necessary resources and learning facilities. The encouragement, understanding, and motivation of family members and peers of this project in the successful completion of this project are thanked.

Declaration

The dissertation presented is an original work done by the author and has not been presented before for any academic award. The sources of information have been cited and listed properly. This is just mere coincidence, and the applicable ethical code of conduct in academic research has been well adhered to during the research.

Abstract

The proposed study will concentrate on designing and testing a machine learning-driven system to predict food waste within food supply chains on the global level. The study solves the increased problem of food waste by examining the country-level data of household, retail, and food service levels. The global food waste dataset was available in a publicly availed set, and as a result, exploratory data analysis, feature engineering, and predictive modelling were used.

The method used for analysis was supervised machine learning, such as the Random Forest regression, Gradient Boosting regression, and Long Short-Term Memory (LSTM) neural networks. The processes of feature scaling, encoding, and systematic training of models were used to assure reliability and reproducibility. Mean absolute error, root mean squared error, and coefficient of determination (R^2) were used to evaluate the model performance.

The findings showed that gradient boosting regression had the best predictive value, with nearly 94 percent variance in terms of food waste per capita being explained. The household waste indicators were the most powerful predictors, with special focus on consumer behavior as the major determinant of food waste production. Also, the percentile-based risk identification mechanism was also useful in identifying countries that are at risk.

The paper finds that machine learning can be useful to facilitate data-driven prediction and prioritization of food waste. The proposed framework has viable importance to policymakers and the supply chain stakeholders since it can facilitate specific interventions and make improved decisions that can lessen the food waste in the world.

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Chapter 1: Introduction

Among the global issues that have posed the most serious challenges to the food security, economic stability, and environmental sustainability of the earth is food waste. A big percentage of the food that is produced throughout the globe is wasted at some point along the chain of supply, such as during production, storage, transportation, processing, and retail distribution (Birkmaier, Imeri and Reiner, 2024). These disastrous losses not only cost resources in terms of water and energy as well as labor; they also contribute a lot to the greenhouse gases. The conventional supply chain management schemes are very dependent on past averages and reactive decision-making, which is usually not adequate to manage the dynamics and uncertainty of food demand and perishability. In this respect, machine learning-driven data-driven models that operate on the basis of advanced analytics present potentially promising practices to predict, trace, and avert food waste (Kayikci*et al.*, 2022). The proposed paper is about the design and implementation of the use of predictive models using real-world food waste records so as to aid in making informed decisions and enhancing supply chain efficiency.

1.1 Background and Rationale

Globalization, diversified consumer preference, and increased quality standards have made the modern food supply chain become complex. As much as the technological enhancement has increased the production scale and the area of distribution, it has also presented new challenges associated with unstable demand and the coordination among the stakeholders (Woolley *et al.*, 2022). The most common causes of food waste are when the demand is lower than supply, the goods become spoiled, or there are problems with logistics and inventory management. Emerging trends in data analytics, machine learning, and cloud computing have helped organizations to gather and crunch huge quantities of supply chain data. The reason behind this research is to use these technologies to utilize raw data to attain actionable insights (Kabadurmus*et al.*, 2023). Through predictive analytics of food waste data, the actors in the supply chain will be able to predict the pattern optimization of inventory and effectual timely interventions to manage operational performance and sustainability agendas.

1.2 Problem Statement

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Supply Chain Stage	Major Causes of Waste	Data Characteristics	Impact Severity
Production	Overproduction, weather uncertainty	Seasonal, volume-based	High
Processing	Quality trimming, sorting losses	Batch-wise records	Medium
Storage	Temperature failure, long holding time	Time-series sensor data	High
Transportation	Delays, poor handling	Route and duration data	Medium
Retail	Demand mismatch, expiry	Sales and inventory data	High

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Supply Chain Level	Waste Drivers	Predictive Focus	Literature Insight
Production	Demand uncertainty	Yield forecasting	Limited integration
Processing	Inefficient operations	Process optimization	Data scarcity
Storage	Cold chain issues	Spoilage prediction	Sensor dependency
Transportation	Transit delays	Risk assessment	Reliability issues
Retail	Overstocking	Demand forecasting	Fragmented models

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Algorithm	Application Area	Data Type	Key Advantage
Random Forest	Inventory optimization	Structured data	Robust to noise
Gradient Boosting	Demand prediction	High-dimensional	High accuracy
LSTM	Temporal waste trends	Time-series	Captures seasonality
Regression Models	Baseline forecasting	Numeric data	Interpretability
Clustering	Waste pattern discovery	Unlabeled data	Insight extraction

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Model Type	Examples	Strengths	Limitations
Statistical	Linear Regression	Simple, interpretable	Poor with nonlinearity
Time-Series	ARIMA	Captures trends	Limited scalability
Machine Learning	Random Forest	High accuracy	Low interpretability
Ensemble Models	Gradient Boosting	Handles complexity	High computation
Deep Learning	LSTM	Temporal learning	Requires large data

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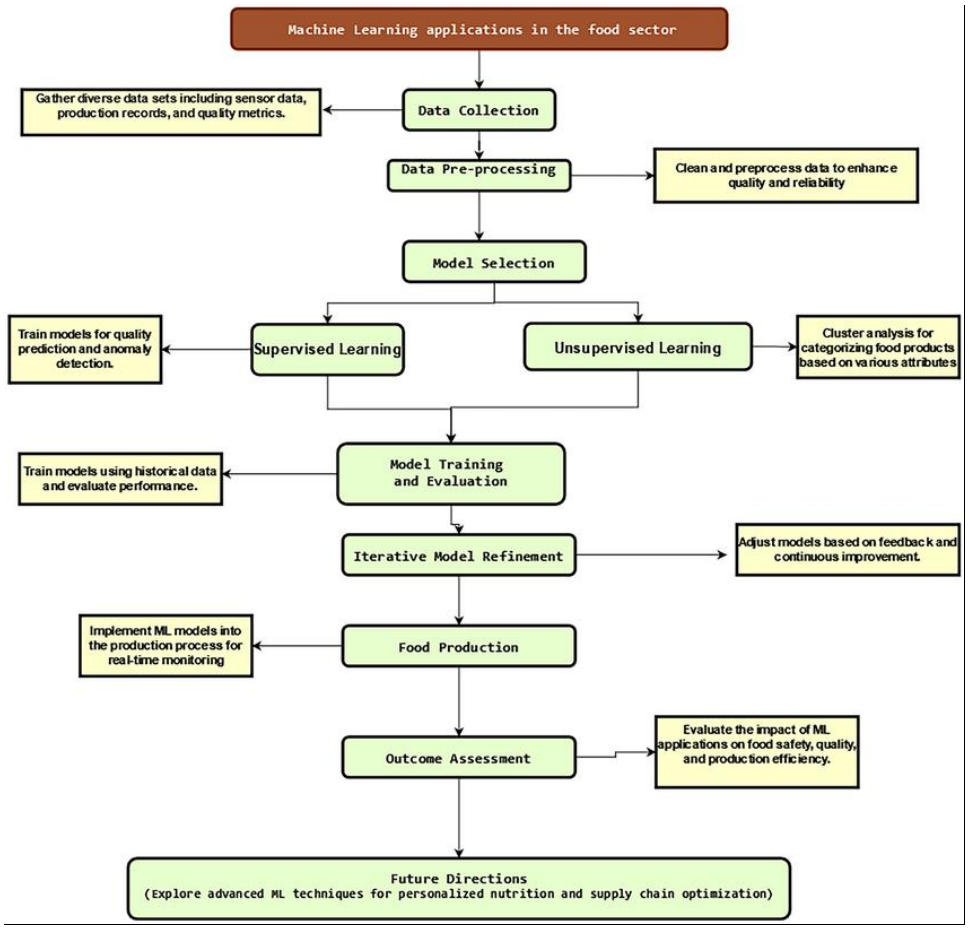
Machine Learning applications in the food sector
Data Collection
Gather diverse data sets including sensor data, production records, and quality metrics.
Data Pre-processing
Clean and preprocess data to enhance quality and reliability
Model Selection
Supervised Learning
Train models for quality prediction and anomaly detection.
Unsupervised Learning
Cluster analysis for categorizing food products based on various attributes
Model Training and Evaluation
Train models using historical data and evaluate performance.
Iterative Model Refinement
Adjust models based on feedback and continuous improvement.
Food Production
Implement ML models into the production process for real-time monitoring
Outcome Assessment
Evaluate the impact of ML applications on food safety, quality, and production efficiency.
Future Directions
(Explore advanced ML techniques for personalized nutrition and supply chain optimization)

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Challenge Type	Description	Impact	Research Gap
Data Quality	Missing and noisy data	Reduced accuracy	Robust preprocessing
Integration	Fragmented systems	Limited scalability	End-to-end models
Interpretability	Black-box models	Low trust	Explainable AI
Implementation	Resource constraints	Low adoption	Practical frameworks
Ethics	Data privacy concerns	Trust issues	Responsible AI

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	Country	combined figures (kg/capita/year)	Household estimate (kg/capita/year)	Household estimate (tonnes/year)	Retail estimate (kg/capita/year)	Retail estimate (tonnes/year)	Food service estimate (kg/capita/year)	Food service estimate (tonnes/year)	Confidence in estimate	M49 code	Region	Sourc
0	Afghanistan	126	82	3109153	16	594982	28	1051783	Very Low Confidence	4	Southern Asia	https://www.unep.org/resources/report/unep foo.
1	Albania	127	83	238492	16	45058	28	79651	Very Low Confidence	8	Southern Europe	https://www.unep.org/resources/report/unep foo.
2	Algeria	135	91	3918529	16	673360	28	1190335	Very Low Confidence	12	Northern Africa	https://www.unep.org/resources/report/unep foo.
3	Andorra	123	84	6497	13	988	26	1971	Low Confidence	20	Southern Europe	https://www.unep.org/resources/report/unep foo.
4	Angola	144	100	3169523	16	497755	28	879908	Very Low Confidence	24	Sub-Saharan Africa	https://www.unep.org/resources/report/unep foo.

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 0   Country                               214 non-null    object
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 2   Household estimate (kg/capita/year)   214 non-null    int64
 3   Household estimate (tonnes/year)      214 non-null    int64
 4   Retail estimate (kg/capita/year)      214 non-null    int64
 5   Retail estimate (tonnes/year)         214 non-null    int64
 6   Food service estimate (kg/capita/year) 214 non-null   int64
 7   Food service estimate (tonnes/year)   214 non-null    int64
 8   Confidence in estimate                214 non-null    object
 9   M49 code                              214 non-null    int64
 10  Region                                214 non-null    object
 11  Source                                214 non-null    object
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memory usage: 20.2+ KB
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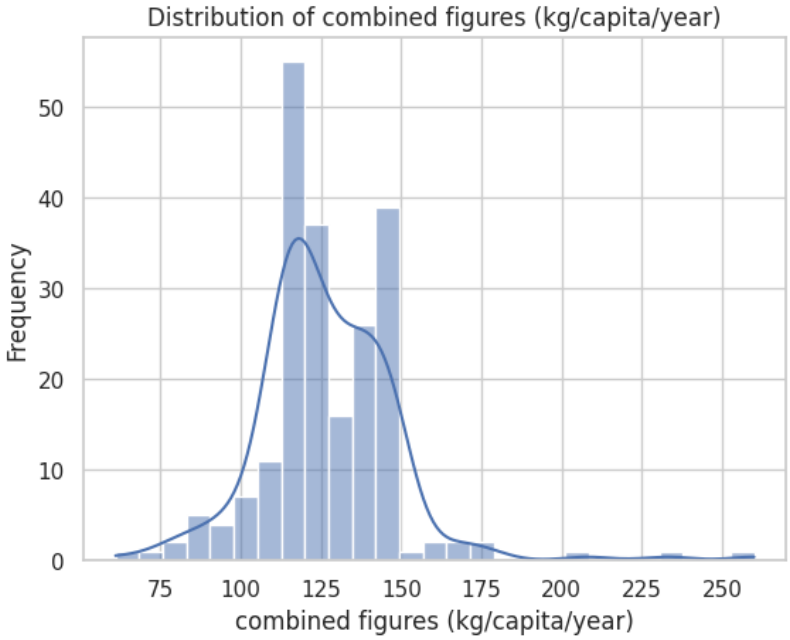

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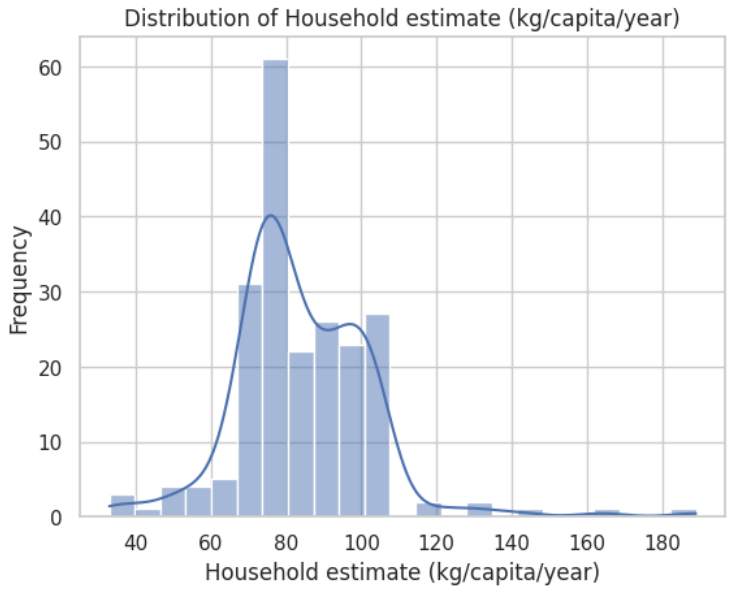

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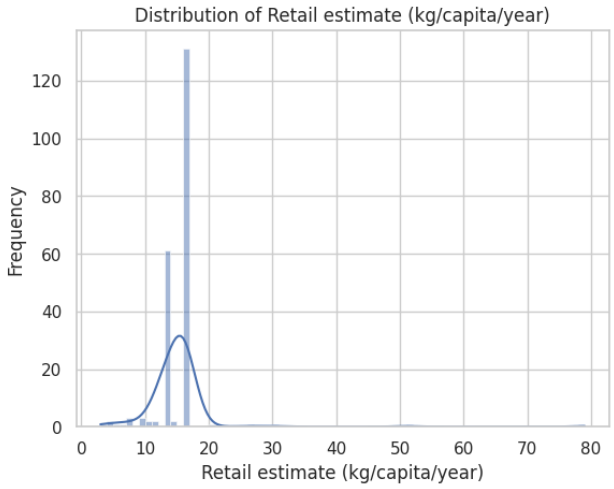

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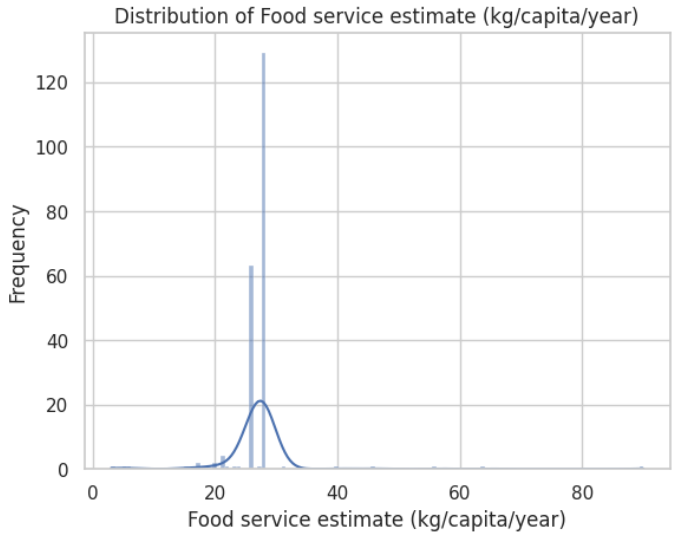

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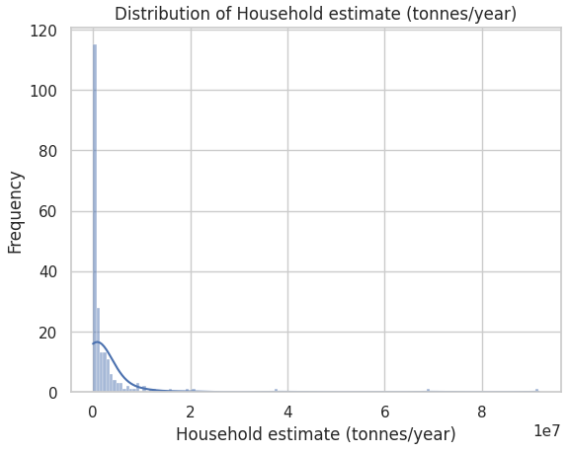

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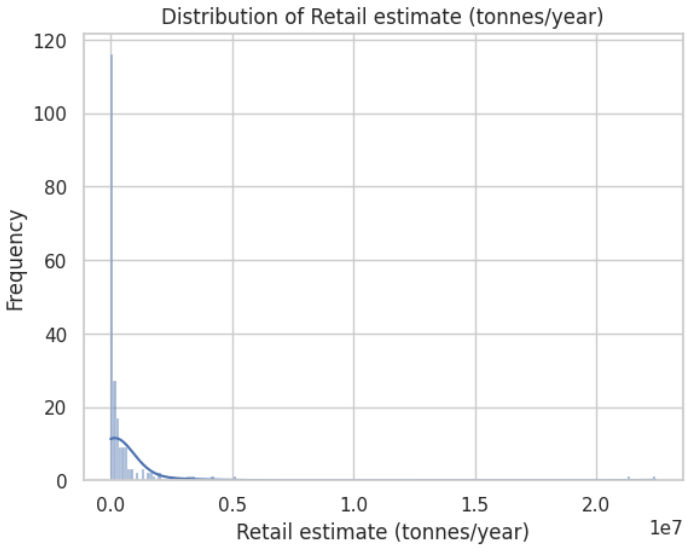
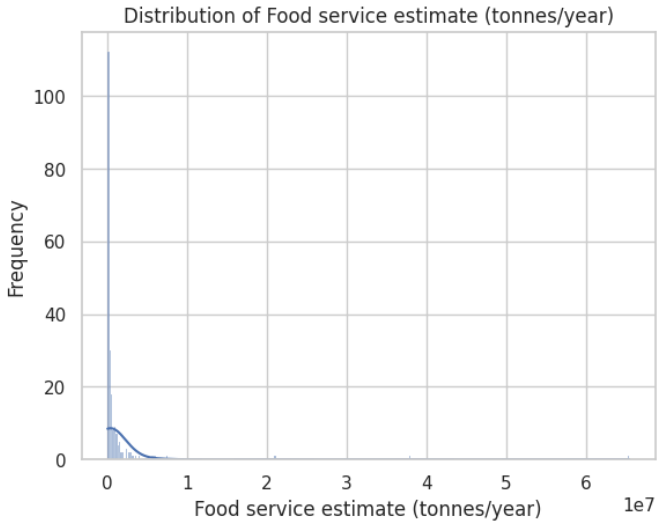

Distribution of Retail estimate (tonnes/year)
Frequency
120
100
80
60
40
20
0
0.0
0.5
1.0
1.5
2.0
1e7
Retail estimate (tonnes/year)



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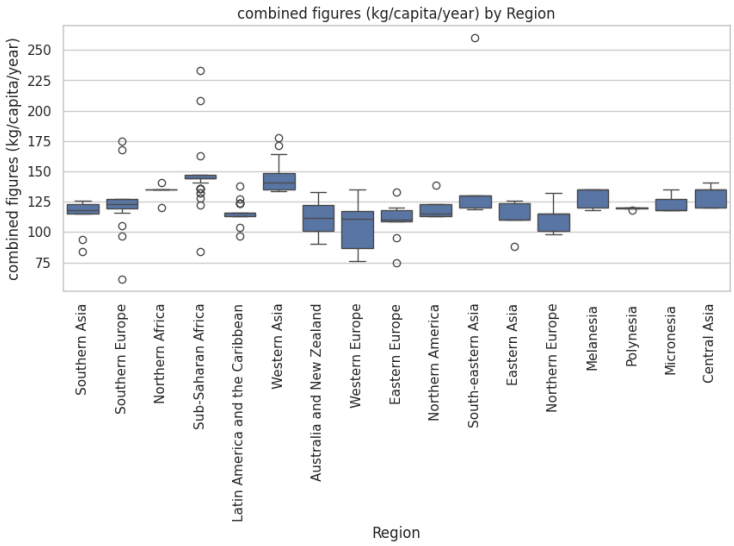

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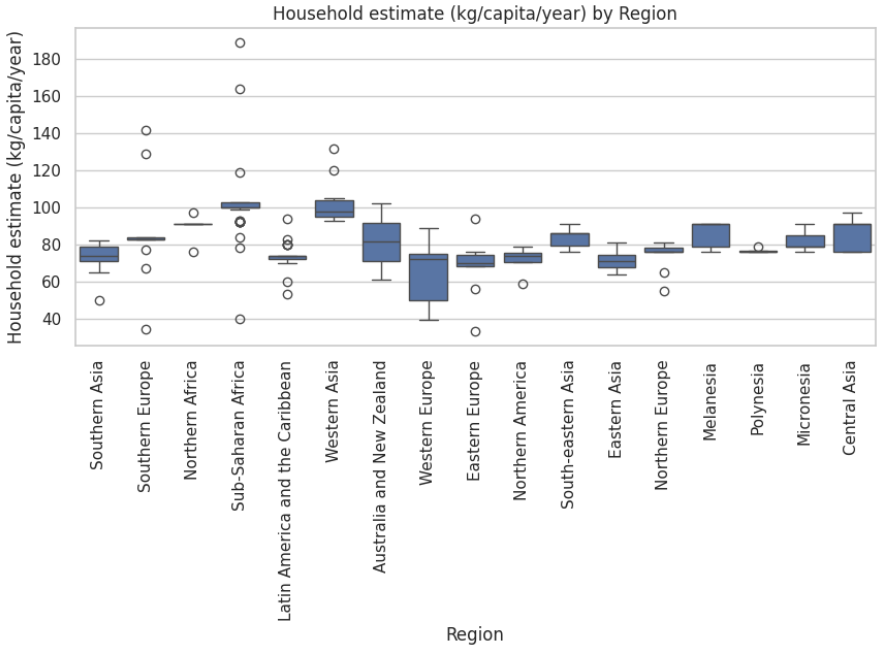
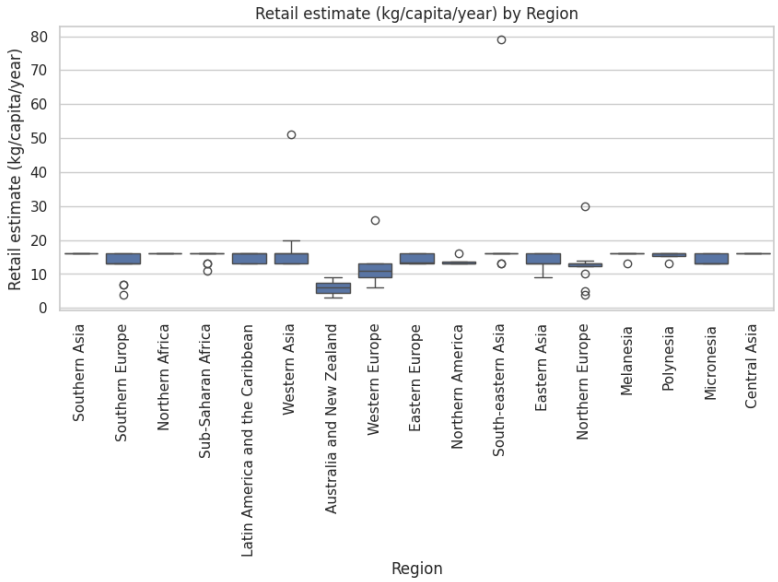

Household estimate (kg/capita/year) by Region
Household estimate (kg/capita/year)
180
160
140
120
100
80
60
40
Southern Asia
Southern Europe
Northern Africa
Sub-Saharan Africa
Latin America and the Caribbean
Western Asia
Australia and New Zealand
Western Europe
Eastern Europe
Northern America
South-eastern Asia
Eastern Asia
Northern Europe
Melanesia
Polynesia
Micronesia
Central Asia
Region



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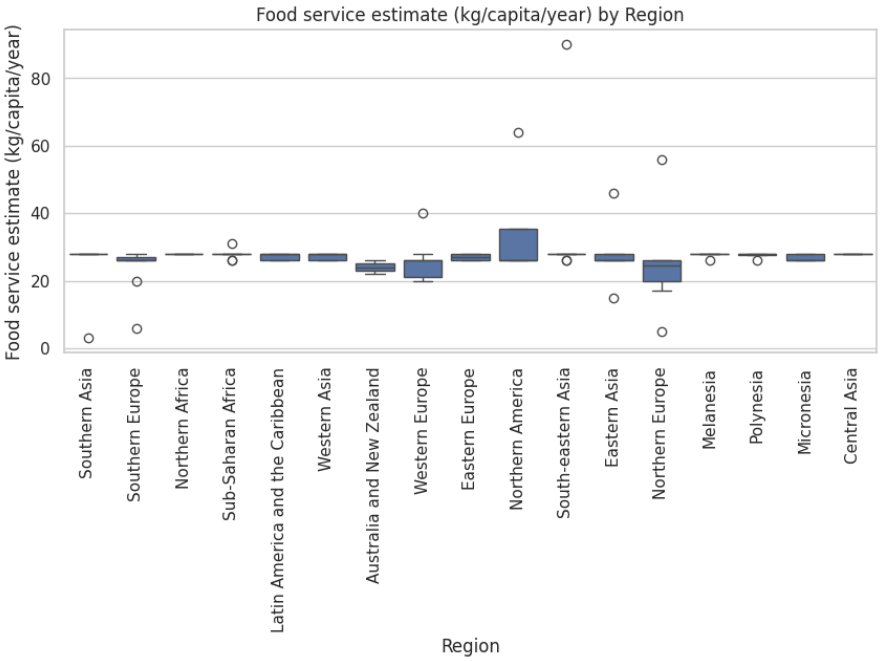

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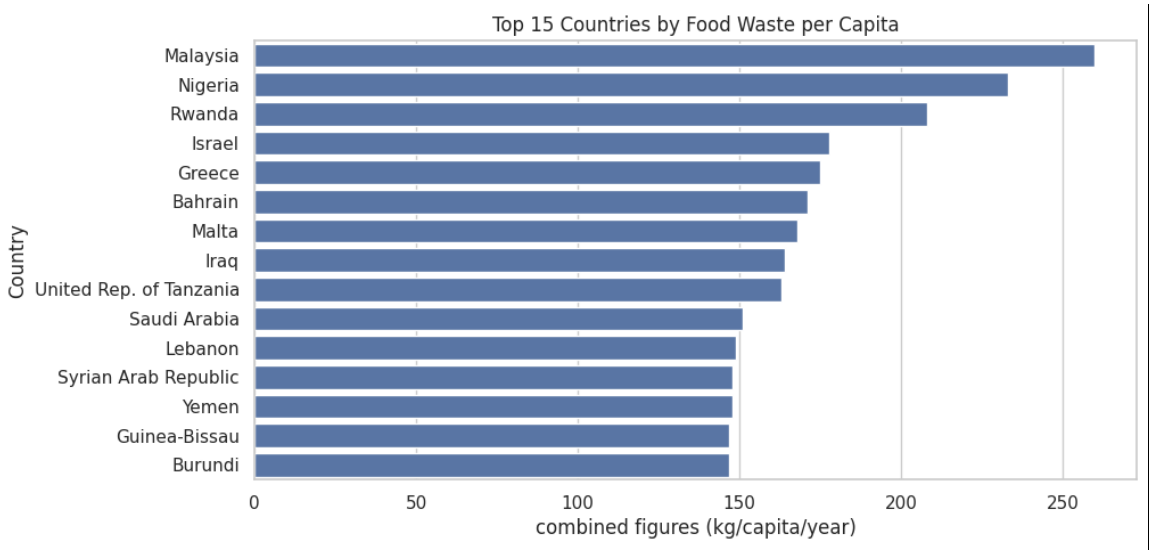

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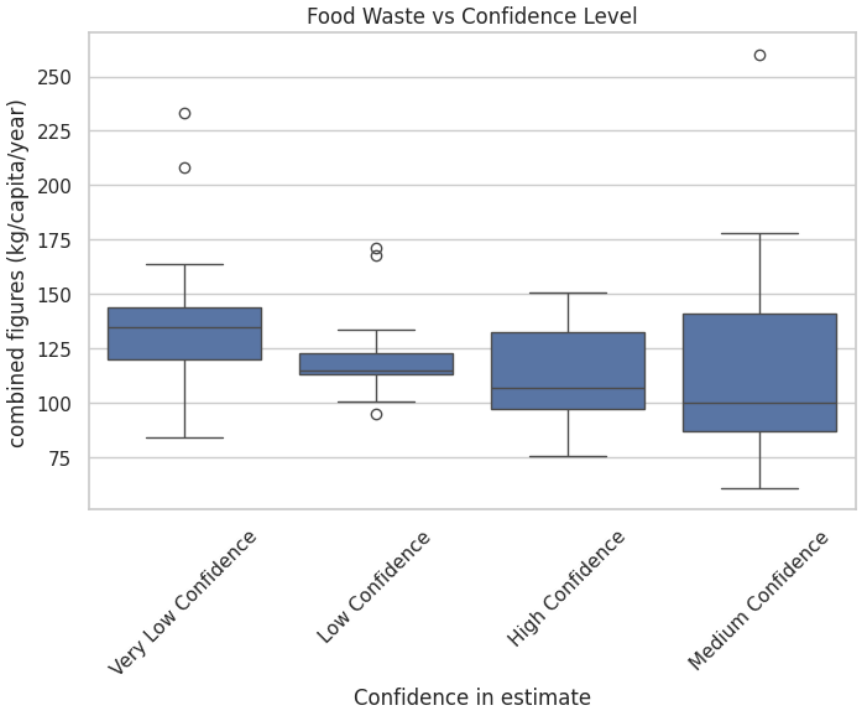
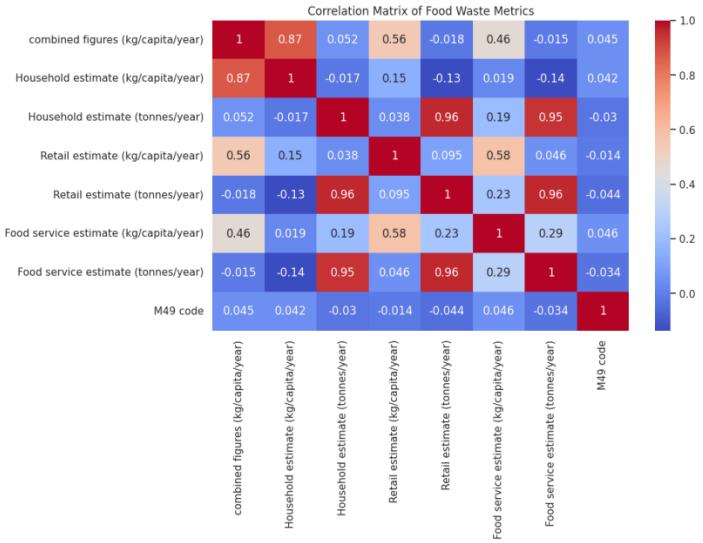

Food Waste vs Confidence Level
combined figures (kg/capita/year)
250
225
200
175
150
125
100
75
Very Low Confidence
Low Confidence
High Confidence
Medium Confidence
Confidence in estimate


Correlation Matrix of Food Waste Metrics
combined figures (kg/capita/year)
Household estimate (kg/capita/year)
Household estimate (tonnes/year)
Retail estimate (kg/capita/year)
Retail estimate (tonnes/year)
Food service estimate (kg/capita/year)
Food service estimate (tonnes/year)
M49 code
1 0.87 0.052 0.56 -0.018 0.46 -0.015 0.045
0.87 1 -0.017 0.15 -0.13 0.019 -0.14 0.042
0.052 -0.017 1 0.038 0.96 0.19 0.95 -0.03
0.56 0.15 0.038 1 0.095 0.58 0.046 -0.014
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0.46 0.019 0.19 0.58 0.23 1 0.29 0.046
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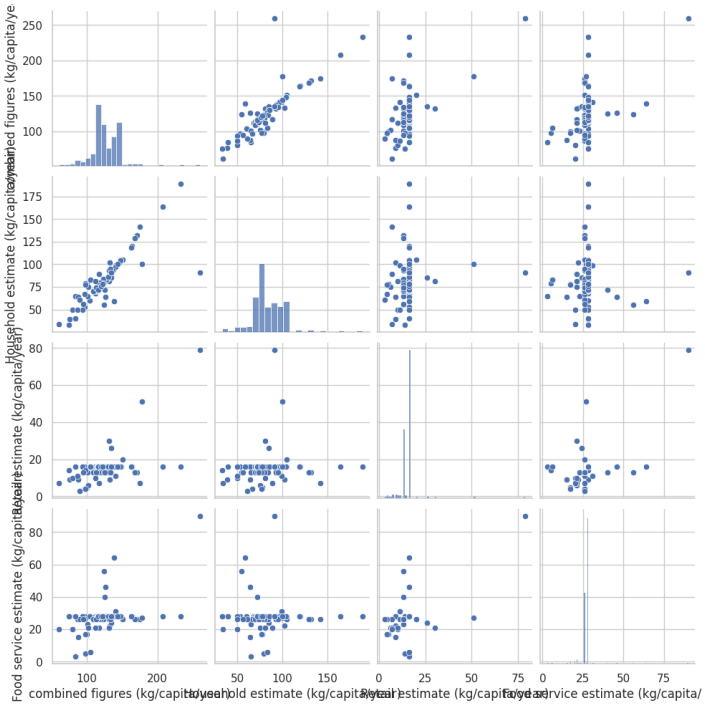

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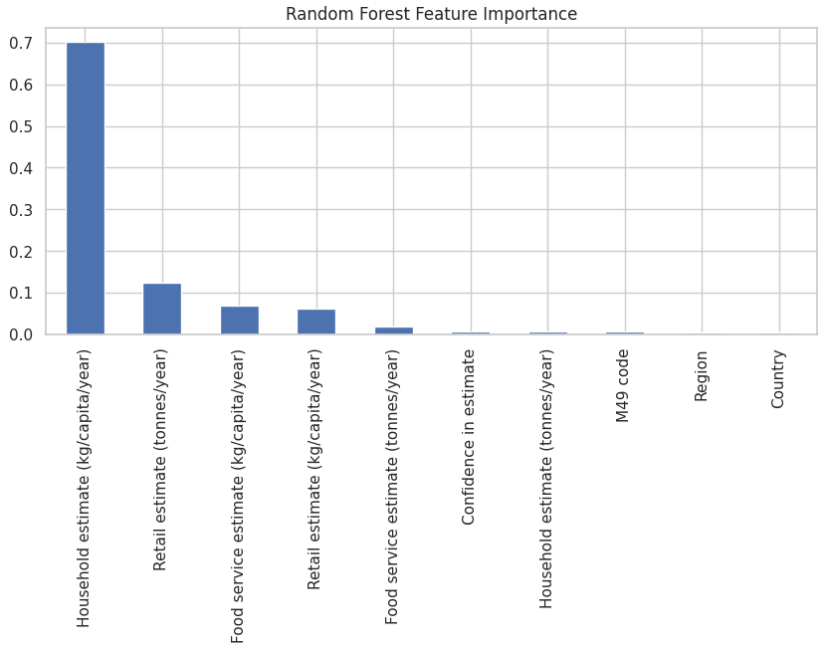

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```python
# Core
import numpy as np
import pandas as pd

# Visualisation
import matplotlib.pyplot as plt
import seaborn as sns

# ML Models
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Time series / Deep Learning
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.optimizers import Adam

# Utilities
import warnings
warnings.filterwarnings("ignore")
```

```python
!pip install kagglehub
```

```python
import kagglehub
import pandas as pd
import os
```

```python
## Download Food Waste Dataset

dataset_path = kagglehub.dataset_download(
    "joebeachcapital/food-waste"
)

print("Dataset downloaded to:", dataset_path)
```

```
Downloading from https://www.kaggle.com/api/v1/datasets/download/joebeachcapital/food-waste?dataset_version_number=1...
100%|██████████| 6.99k/6.99k [00:00<00:00, 6.48MB/s]
Extracting files...
Dataset downloaded to: /root/.cache/kagglehub/datasets/joebeachcapital/food-waste/versions/1
```

```python
# List all files in the dataset directory
os.listdir(dataset_path)
```

```
['Food Waste data and research - by country.csv']
```

```python
file_path = os.path.join(dataset_path, "Food Waste data and research - by country.csv")
```

```python
## Distribution of All Waste Metrics (kg/capita/year)

metrics_kg = [
    "combined figures (kg/capita/year)",
    "Household estimate (kg/capita/year)",
    "Retail estimate (kg/capita/year)",
    "Food service estimate (kg/capita/year)"
]

for col in metrics_kg:
    plt.figure()
    sns.histplot(df[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```

```python
## Distribution of All Waste Metrics (tonnes/year)

metrics_tonnes = [
    "Household estimate (tonnes/year)",
    "Retail estimate (tonnes/year)",
    "Food service estimate (tonnes/year)"
]

for col in metrics_tonnes:
    plt.figure()
    sns.histplot(df[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```

```python
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```python
df_model = df.copy()

# Encode categorical variables
le = LabelEncoder()
df_model["Country"] = le.fit_transform(df_model["Country"])
df_model["Region"] = le.fit_transform(df_model["Region"])
df_model["Confidence in estimate"] = le.fit_transform(df_model["Confidence in estimate"])
```

```python
## Define Features and Target

X = df_model.drop(
    columns=[
        "combined figures (kg/capita/year)",
        "Source"
    ]
)

y = df_model["combined figures (kg/capita/year)"]
```

```python
scaler = StandardScaler()
```

```python
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)

rf = RandomForestRegressor(
    n_estimators=300,
    random_state=42
)

rf.fit(X_train, y_train)
rf_preds = rf.predict(X_test)

print("Random Forest MAE:", mean_absolute_error(y_test, rf_preds))
print("Random Forest RMSE:", mean_squared_error(y_test, rf_preds))
print("Random Forest R²:", r2_score(y_test, rf_preds))
```

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```python
from sklearn.ensemble import GradientBoostingRegressor
gb = GradientBoostingRegressor(
    n_estimators=300,
    learning_rate=0.05,
    random_state=42
)

gb.fit(X_train, y_train)
gb_preds = gb.predict(X_test)

print("Gradient Boosting MAE:", mean_absolute_error(y_test, gb_preds))
print("Gradient Boosting RMSE:", mean_squared_error(y_test, gb_preds))
print("Gradient Boosting R²:", r2_score(y_test, gb_preds))
```

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```python
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.optimizers import Adam


df_ts = df.sort_values("M49 code")

series = df_ts["combined figures (kg/capita/year)"].values

def create_sequences(data, seq_len=5):
    X, y = [], []
    for i in range(len(data) - seq_len):
        X.append(data[i:i+seq_len])
        y.append(data[i+seq_len])
    return np.array(X), np.array(y)

X_seq, y_seq = create_sequences(series)
X_seq = X_seq.reshape((X_seq.shape[0], X_seq.shape[1], 1))


lstm_model = Sequential([
    LSTM(64, return_sequences=True, input_shape=(X_seq.shape[1], 1)),
    Dropout(0.2),
    LSTM(32),
    Dense(1)
])

lstm_model.compile(
    optimizer=Adam(learning_rate=0.001),
    loss="mse"
)

lstm_model.summary()
```

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```
plt.figure(figsize=(10,4))
plt.plot(y_test.values[:50], label="Actual")
plt.plot(rf_preds[:50], label="Random Forest")
plt.plot(gb_preds[:50], label="Gradient Boosting")
plt.legend()
plt.title("Model Prediction Comparison")
plt.show()
```

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```python
results = df.iloc[y_test.index].copy()
results["Predicted Waste"] = rf_preds

high_risk = results[
    results["Predicted Waste"] >
    results["Predicted Waste"].quantile(0.75)
]

high_risk[["Country", "Predicted Waste"]].head()
```

	Country	Predicted Waste
9	Australia	157.683333
191	Togo	147.040000
182	State of Palestine	144.933333
73	Georgia	145.230000
167	Saudi Arabia	158.636667

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```
## boxplots by Region (kg/capita/year)

for col in metrics_kg:
    plt.figure(figsize=(10,4))
    sns.boxplot(x="Region", y=col, data=df)
    plt.xticks(rotation=90)
    plt.title(f"{col} by Region")
    plt.show()
```

```
numeric_cols = df.select_dtypes(include="int64")

corr_matrix = numeric_cols.corr()

plt.figure(figsize=(10,6))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix of Food Waste Metrics")
plt.show()
```

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```python
## Pairwise Relationships (Key Variables)

sns.pairplot(
    df,
    vars=[
        "combined figures (kg/capita/year)",
        "Household estimate (kg/capita/year)",
        "Retail estimate (kg/capita/year)",
        "Food service estimate (kg/capita/year)"
    ]
)
plt.show()
```

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```python
df_model = df.copy()

# Encode categorical variables
le = LabelEncoder()
df_model["Country"] = le.fit_transform(df_model["Country"])
df_model["Region"] = le.fit_transform(df_model["Region"])
df_model["Confidence in estimate"] = le.fit_transform(df_model["Confidence in estimate"])
```

```python
## Define Features and Target

X = df_model.drop(
    columns=[
        "combined figures (kg/capita/year)",
        "Source"
    ]
)

y = df_model["combined figures (kg/capita/year)"]
```

```python
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```python
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)

rf = RandomForestRegressor(
    n_estimators=300,
    random_state=42
)

rf.fit(X_train, y_train)
rf_preds = rf.predict(X_test)

print("Random Forest MAE:", mean_absolute_error(y_test, rf_preds))
print("Random Forest RMSE:", mean_squared_error(y_test, rf_preds))
print("Random Forest R²:", r2_score(y_test, rf_preds))
```

```
Random Forest MAE: 4.104418604651161
Random Forest RMSE: 70.23696046511627
Random Forest R²: 0.8381894956852407
```

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```python
from sklearn.ensemble import GradientBoostingRegressor
gb = GradientBoostingRegressor(
    n_estimators=300,
    learning_rate=0.05,
    random_state=42
)

gb.fit(X_train, y_train)
gb_preds = gb.predict(X_test)

print("Gradient Boosting MAE:", mean_absolute_error(y_test, gb_preds))
print("Gradient Boosting RMSE:", mean_squared_error(y_test, gb_preds))
print("Gradient Boosting R²:", r2_score(y_test, gb_preds))
```

```
Gradient Boosting MAE: 2.227126916836136
Gradient Boosting RMSE: 26.93258117642615
Gradient Boosting R²: 0.9379532583158958
```

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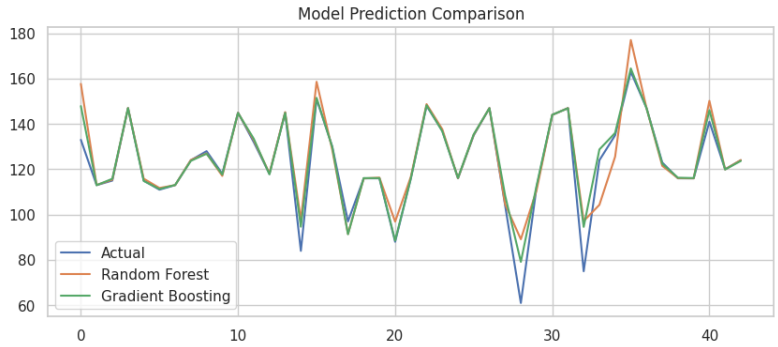

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```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 5, 64)	16,896
dropout (Dropout)	(None, 5, 64)	0
lstm_1 (LSTM)	(None, 32)	12,416
dense (Dense)	(None, 1)	33

```
Total params: 29,345 (114.63 KB)

Trainable params: 29,345 (114.63 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/50
11/11 ━━━━━━━━━━━━━━━━━━━━ 4s 53ms/step - loss: 17039.4160 - val_loss: 15793.6895
Epoch 2/50
11/11 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step - loss: 16030.5312 - val_loss: 15381.0459
Epoch 3/50
11/11 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 15507.8916 - val_loss: 14962.3896
Epoch 4/50
11/11 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 14912.4922 - val_loss: 14591.3125
Epoch 5/50
11/11 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 14658.8799 - val_loss: 14329.6816
```

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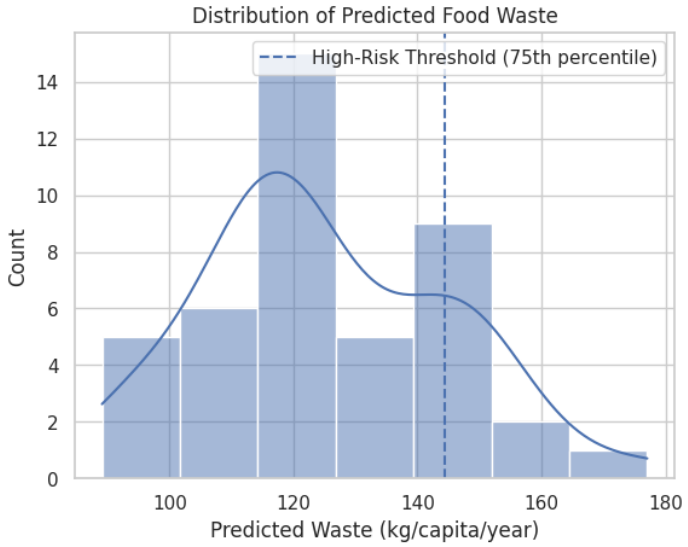

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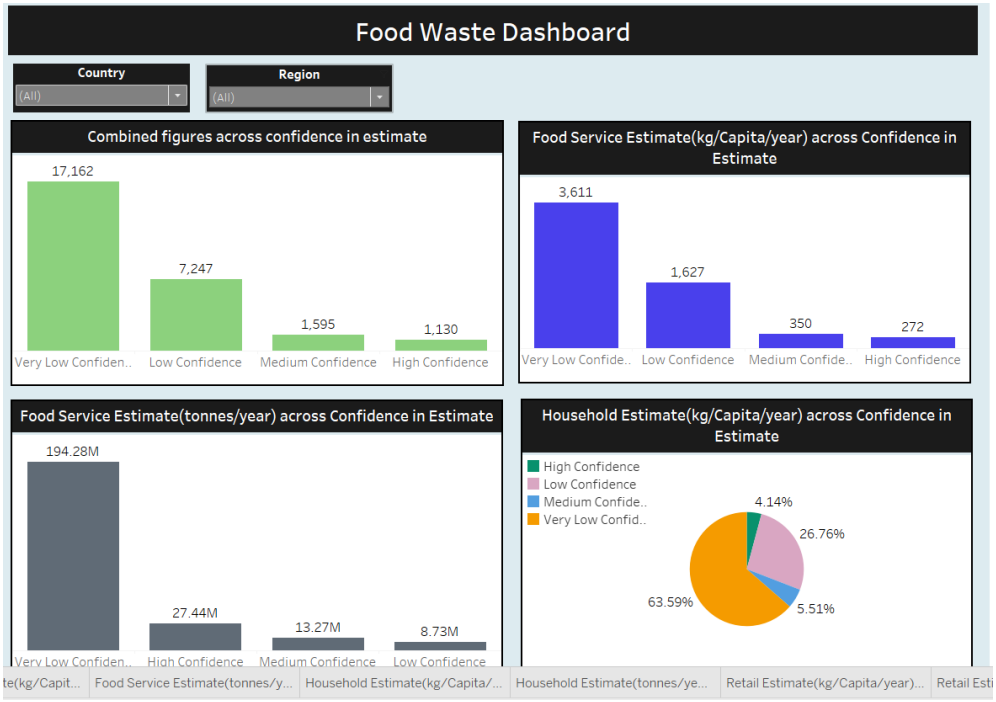
Food Waste Dashboard
Country
(All)
Region
(All)
Combined figures across confidence in estimate
17,162
7,247
1,595
1,130
Very Low Confiden..
Low Confidence
Medium Confidence
High Confidence
Food Service Estimate(kg/Capita/year) across Confidence in Estimate
3,611
1,627
350
272
Very Low Confide..
Low Confidence
Medium Confide..
High Confidence
Food Service Estimate(tonnes/year) across Confidence in Estimate
194.28M
27.44M
13.27M
8.73M
Very Low Confiden..
High Confidence
Medium Confidence
Low Confidence
Household Estimate(kg/Capita/year) across Confidence in Estimate
High Confidence
Low Confidence
Medium Confide..
Very Low Confid..
4.14%
26.76%
63.59%
5.51%
te(kg/Capit...
Food Service Estimate(tonnes/y...
Household Estimate(kg/Capita/...
Household Estimate(tonnes/ye...
Retail Estimate(kg/Capita/year)...
Retail Esti

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